A/C COMPRESSOR CLUTCH (Continued)

tion for testing of the compressor clutch circuit and PCM control. The following components must be checked and repaired as required before you can complete testing of the clutch coil:

- Fuses in the fuse block and the power distribution center (PDC)
- A/C-heater mode control
- A/C clutch relay
- A/C High pressure switch (4.0L engine)
- A/C Low pressure switch
- Powertrain control module (PCM)

(5) The A/C compressor clutch coil is acceptable if the current draw is 2.0 to 3.7 amperes. Specifications apply for a work area temperature of 21° C (70° F). If voltage is more than 12.5 volts, add electrical loads by turning on electrical accessories until voltage reads below 12.5 volts.

(a) If the A/C clutch coil current reading is zero, the coil is open and must be replaced.

(b) If the A/C clutch coil current reading is above specifications, the coil is shorted and must be replaced.

STANDARD PROCEDURE

DIODE REPLACEMENT

(1) Disconnect and isolate the negative battery cable.

(2) Locate the diode in the A/C clutch wire harness and remove the protective covering.

(3) Remove the diode from the wire harness while paying attention to the current flow direction (Fig. 2).

(4) Remove the insulation from the wires in the harness. Only remove enough insulation to solder in the new diode.

(5) Install the new diode in the harness, making sure current flow is correct. If necessary refer to the appropriate wiring diagram for current flow.

CAUTION: To prevent damage to the diode and wiring, DO NOT use acid core solder.

(6) Solder the connections together using rosin core type solder only.

(7) Tape the diode to the harness using electrical tape making, sure the diode is completely sealed from the elements.

(8) Reconnect the negative battery cable, and test A/C clutch operation.

A/C CLUTCH INSPECTION

NOTE: The A/C clutch can be serviced in the vehicle. The refrigerant system can remain fully-charged during compressor clutch, pulley and bearing assembly, or coil replacement.

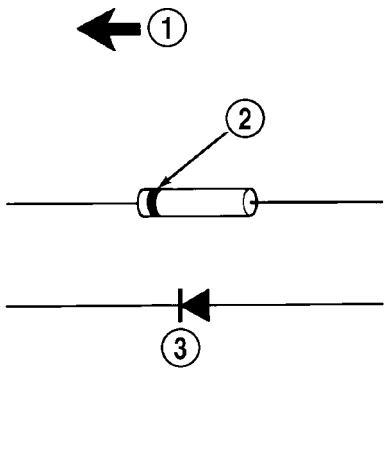

Fig. 2 Diode Identification

1 - CURRENT FLOW
2 - BAND INDICATES CURRENT FLOW
3 - DIODE AS SHOWN IN WIRE DIAGRAMS

Examine the friction surfaces of the clutch pulley and the clutch plate for wear (Fig. 3). The pulley and plate should be replaced if there is excessive wear or scoring.

If the friction surfaces are oily, inspect the shaft and nose area of the A/C compressor for refrigerant oil. If refrigerant oil is found, the compressor shaft seal is leaking and the A/C compressor must be replaced.

Check the clutch pulley bearing for roughness or excessive leakage of grease. Replace the pulley and bearing assembly, if required.

A/C CLUTCH BREAK-IN

After a new A/C compressor clutch has been installed, cycle the compressor clutch approximately twenty times (five seconds on, then five seconds off). During this procedure, set the A/C-heater controls to the A/C Recirculation Mode, the blower motor in the highest speed position, and the engine speed at 1500 to 2000 rpm. This procedure (burnishing) will seat the opposing friction surfaces and provide a higher compressor clutch torque capability.

REMOVAL

NOTE: The compressor clutch can be serviced in the vehicle. The refrigerant system can remain fully-charged during compressor clutch, pulley, or coil replacement.

(1) Disconnect and isolate the battery negative cable.

(2) Remove the serpentine drive belt (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - REMOVAL) and (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - REMOVAL).